Chapter 1

Achievement in Engineering and Technology Study

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1.1. Introduction

According to experts, success in any task or field begins with a positive mindset and the belief that **"I can do it!"** However, confidence alone is not enough. To turn that belief into achievement, you need to set clear goals and consistently practice.

Goals give you something to work toward and a way to measure your progress. They provide direction, motivation, and purpose, helping you stay focused even when challenges arise. In many ways, goals are your **internal compass**—they

guide your decisions, give energy to your efforts, and help you understand where you want to go.

Whether you are working toward a small accomplishment or pursuing a big dream, setting meaningful goals and taking consistent action can help you create the future you want. **Believe in yourself, set your goals, practice, and keep moving forward.**

Practice gives you your limit, the drive to practice is like watering a seed, it's how growth happens, whether you're building skills, mastering a craft, or improving your mindset. Practice improves precision, accuracy, timing, and efficiency. It turns effort into elegance, reinforces understanding, whether it's math or meditation, regular practice deepens comprehension by connecting ideas through repetition. Strengthens resilience: You face mistakes, frustrations, plateaus—and learn to push through. It's mental training as much as physical. Unlocks mastery: With time, practice transforms potential into excellence. It's the difference between dabbling and dominating. It's not just about doing something over and over, it's about intentional repetition with a goal. Like sculpting, each swing of the chisel refines the shape.

1.2. Success in Engineering Technology

To be successful in Engineering Technology study you need to strengthen your commitment to your goals and practice, create a model to follow for your education, and structure your life situation. **Success in Engineering Technology** isn't just about acting on tests, it's about learning how to think like a problem-solver, a builder, and a visionary.

Here's a roadmap that can help you thrive in your studies:

1. **Master the Fundamentals**
   - Get rock-solid in math (algebra, trigonometry, calculus) and physics—they're the language of engineering.
   - Understand the basics of electronics, mechanics, computer science, and materials science depending on your specialization.

2. **Hands-On Learning**

- Engineering tech is all about application. Work on real projects, labs, and prototypes as much as possible.
- Don't just read about circuits—build one. Don't just study CAD software—design something with it.

3. **Collaborate with Peers**

- Join study groups, hackathons, or engineering clubs.
- Real-world engineering is collaborative. Practicing teamwork early helps you build communication and leadership skills.

4. **Use Resources Wisely**

- Explore online tutorials, YouTube channels, textbooks, and professors' office hours.
- Get comfy with software tools like MATLAB, SolidWorks, AutoCAD, or whatever's relevant to your field.

5. **Iterate and Reflect**

- Treat mistakes as stepping stones. Every design that fails teaches you something new.
- Keep track of your projects and what you learned from each—kind of like your own engineering journal.

6. **Stay Curious**

- Engineering is constantly evolving. Follow innovations in robotics, renewable energy, AI, or sustainable design—whatever excites you.
- Subscribe to engineering magazines or online forums like IEEE Spectrum or TechCrunch Engineering.

Class Activity 1

- Write your personal and professional goals.
- What Strategies will you use for strengthening your commitment

Strategies for strengthening your commitment

- Clarifying your goals
- Learning as much as you can about engineering-research four different engineering fields and writing about the courses they need to take and the type of jobs available for them.
- Prepare a roadmap for your study.

Bases to Success in Engineering Study

- Effort – "Work Hard"
- Approach – "Work Smart"
- Attitude – "Think Positively"

Setting a Model for Your Education- As you go through your study at any college you should always be thinking about the following questions and have an efficient answer for each.

Attributes Model – What knowledge, skills, and attitudes will you have when you graduate?

These models will assist you in answering such important questions as:

- What is the purpose of my education?
- What should I know when I graduate?
- How do I know if I am getting an excellent education?
- How can I enhance the quality of my education?
- Will I have the knowledge and skills to get a job and be a successful engineering professional?

Employment Model – What do employers look for in new engineering graduates?

Student Involvement Model – What can you do to ensure that you get a quality education?

Here are a few things describing the importance of student involvement in college.

- Time and energy devoted to studying
- Time spent on campus
- Participation in student organizations
- Interaction with faculty members
- Interaction with other students
- Structure Your Life Situation
- Living Arrangements
- Part-Time Work
- Influence of Family
- Influence of Friends
- Other influences?

Class Activity 2

1. Search for two of your desired jobs after graduation from QCC or a four-year college.
2. Read the job description and write a statement of:
   - Was the job description as you imagined it for the profession that you search for? Explain your answer.
   - After reading the job description, what would be your plan to develop your professional career?

1.3. What is an Engineer and an Engineering Technologist?

What is Engineering- The profession in which math and science - gained by study, experience, and practice - is applied with judgment to develop ways to economically use the materials and forces of nature for the benefit of society?

Engineering is the art and science of solving problems using creativity, math, and technical knowledge. It's all about transforming ideas into real-world solutions that improve how we live, work, and interact with the world around us.

The Engineering Mindset

Being an engineer isn't just about technical skills. It's a way of thinking:

- Break big problems into smaller, solvable ones.
- Test ideas through hands-on experimentation.
- Stay curious and always ask: "How can this be better?"

What is Engineering Technology?

An engineering technologist is a specialist dedicated to apply basic engineering principles and technical skills (hands-on) to develop and implement existing technology.

Engineering Technology is where practical application meets engineering theory. It's a hands-on, problem-solving field that focuses on implementing, operating, and improving systems and technologies designed by engineers.

What is the difference between engineering and engineering technology?

Engineering programs focus on theory and design, ET programs specialize in application and implementation. It follows that engineering programs have higher-level math and theoretical science in their curricula and engineering technology programs tend to put greater emphasis on hands-on laboratory skills.

How It Differs from Traditional Engineering:

Aspect: Focus

- **Engineering:** Conceptual design & theory
- **Engineering Technology:** Applied design & implementation

Aspect: Approach

- **Engineering:** Research and development
- **Engineering Technology:** Operation and optimization

Aspect: Education

- **Engineering:** Heavy on advanced math & science
- **Engineering Technology:** Emphasis on technical skills & labs

Aspect: Career Paths

- **Engineering:** Design engineer, R&D specialist
- **Engineering Technology:** Technician, technologist, project manager

1.4. Who is a Technician, Technologist or Engineer?

Technicians

In general, technicians have a two year degree such as A.A.S (Associate of Applied Science) or A.S.E.T. (Applied Science and Engineering Technology) Technicians with an associate degree can transfer to a bachelor of technology program and complete the bachelor's degree in four years Program College. However, associate degree in engineering technology is not able to have smooth transfer into engineering degree program without losing many credits.

Technicians are most often employed in service jobs. The work typically involves equipment installation, troubleshooting and repair, testing and measuring, maintenance and adjustment, manufacturing, or operation. Technicians in these positions are sometimes called field service technicians, field service engineers, or customer representatives. Technicians also work as engineering technicians, lab technicians, engineering assistant, or associate engineer.

Technologist

Technologists are employed in a large and wide-array of industries, including manufacturing, construction, industrial, maintenance, and management. They may be hired as managers of technology, depending on the technologist's educational emphasis on management preparation. Entry-level positions such as product design, testing, development, systems development, field engineering, technical

operations, and quality control are all common positions for engineering technology graduates.

Engineering Technologist

An engineering technologist is a specialist devoted to the implementation of existing technology within a field of engineering. Technologists often work with engineers in a wide variety of projects by applying basic engineering principles and technical skills. The work of technologists is usually focused on the portion of the technological spectrum closest to product improvement, manufacturing, construction, and engineering operational functions.

Engineers

Engineers design equipment and systems.

They have B.S., M.S., or Ph.D. in Electrical, Mechanical, Civil or other engineering fields.

Engineering Disciplines

- Electrical Engineers

  Work with electricity, electromagnetic fields, and light. Design products such as cell phones, computers, antennas, televisions, and appliances. Design power generation and distribution systems.

- Mechanical Engineers

  Work with machines and systems that involve the production and use of heat

  Design cars, heating and air conditioning systems, and manufacturing systems.

- Computer Engineers

  Research, design, and develop computer hardware and equipment, including chips, analog sensors, circuit. Boards, keyboards, modems, routers and printers.

- Civil Engineers

  Work with physics and materials Design buildings, roads, bridges, storm water systems

- Chemical Engineers

  Work with physics, chemistry and biology. Design medicines, fertilizers, semiconductors, food additives (such as flavorings)

- Aerospace Engineers

  Work with aerodynamics, propulsion and navigation, Design aircraft, spacecraft, satellites, missiles, and rockets.

There are many more disciplines!

From A.A.S to Bachelor in Technology, B.T., degree. To four-year college, B.T. (All or most credit accepted) - The following local colleges have Technology program you can transfer to:

- NYCCT (CUNY)
- Farmingdale State College (SUNY)
- NYIT (Private)
- Vaughn College (Private)

From A.A.S to Bachelor in Science, B.S., degree in Engineering. To four-year college, B.S. (Only transfer liberal art and English courses: physics, math, and ET courses are not transferable in some cases). The following local colleges have Engineering program you can transfer to:

- City College (CUNY)
- Stony Brook University (SUNY)
- Buffalo University (SUNY)
- New York University (Private)
- Hofstra University (Private)
- NYIT (Private)

Class Activity 3

1. Search 3 of the colleges above for their technology and engineering programs and list 4 courses that include the same tops for the course but has different course name in technology and engineering courses only?
2. What is the tuition for the college above?
3. What does it take to be an Engineer?

You don't have to be a genius to be an Engineer? People who become Engineers:

- are CURIOUS about how things work, solve interesting life problems, design and experiment.
- perform calculation using math and science.
- work with others as a team commitment

Do they sound like you? Then you are the right candidate to become a successful engineer.

1.5. Why be an Engineer?

Because it's a fun and creative career with a wide variety of work, countless job opportunities, and the chance to make a real impact by building and improving the world—just ask famous engineers like:

- musician Tom Scholz (Founder and lead singer of the band Boston)
- President Jimmy Carter (39th President, Graduated from the Naval Academy with a Bachelor's Degree in nuclear engineering. Served on a submarine, as a Navy Officer operating the nuclear power plant.)
- comedian Rowan Atkinson (AKA Mr. Bean, Graduated from Newcastle University with a Bachelor's Degree in electrical engineering, then obtained his Masters in engineering from Oxford)
- talk show host Montel Williams (Graduated from the Naval Academy in 1980 with a Bachelor of Science in engineering. Served 9 years as a Navy Officer), who all used their engineering backgrounds in unique and inspiring ways.

How do you become a Technologist or Engineer?

- EDUCATION- College, most engineering degrees are completed in 4-5 years.
- Attend a school that is Accredited by Accreditation Board for Engineering and Technology (ABET)!
- Assures graduates from the program are prepared for graduate school and the workforce
- Degrees from non-accredited schools will preclude licensure, most graduate programs require Bachelor degrees from ABET schools.

How do you become a Technologist or Engineer?

- Join Engineering Honor Societies and participate.
- Work hard to obtain the highest Grade Point Average – especially in your discipline!

How do you become a professional in your field?

- STAY IN SCHOOL and ENJOY EDUCATION! If it was easy, everyone would be an engineer.
- DO WELL IN SCHOOL!
- GRADUATE!

1.6. Engineer Licensure

So, you've earned your engineering degree and are ready to take on the world—but not so fast! In certain engineering disciplines, particularly those that impact public health and safety like civil and structural engineering, obtaining a professional license is essential. To become a licensed Professional Engineer (PE), you must first earn a bachelor's degree from an ABET-accredited program, then gain at least four years of experience under the supervision of a licensed PE, and pass two rigorous competency exams. Even after licensure, becoming a true professional means continuously studying, practicing new skills, and staying current with emerging technologies in your field.

Class Activity 4

After recognizing the difference between a technician, technologist, and engineer, what career would you like to focus on and why?

1. Would you like to continue your career and pursuit a bachelor's degree? Explain.
2. In case you have plans to continue in a 4-year college, which college/university would you like to transfer to and why? If you don't have a college/university in mind, you can search.

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